Anush Ramesh

AI/ML | Computer Vision | Data Science

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Education

Amrita Vishwa Vidyapeetham — Coimbatore, Tamil Nadu
Bachelor of Technology in Electrical and Computer Engineering — *Aug. 2023 – April 2027*

Experience

AI/ML Intern — June 2025 – July 2025
Edunet Foundation

- Completed a 4-week AICTE Microsoft internship focused on AI, ML, and Azure Cloud.
- Built a supervised ML model in Python for cricket match outcome prediction using data preprocessing, feature engineering, and model evaluation.

AI & Edge Computing Intern — June 2026
Larsen & Toubro — *Chennai, India*

- Developed an AI-based Biomedical Waste Segregation Monitoring System for automated biomedical waste classification using Computer Vision techniques.
- Built and annotated a biomedical waste dataset comprising 9,124 images across 8 waste categories using Roboflow.
- Trained, tested, and evaluated a custom YOLOv11 Nano object detection model for biomedical waste detection and performance validation.
- Presented project methodology, implementation, and results to industry mentors.

Projects

Biomedical Waste Segregation Monitoring System | *Python, YOLOv11, OpenCV, Streamlit, Power BI*

- Developed an end-to-end Computer Vision and Object Detection system for automated biomedical waste classification and disposal-bin recommendation.
- Built and annotated a dataset of 9,124 images across 8 biomedical waste classes using Roboflow.
- Achieved 95.5% mAP@50, 91.4% Precision, and 95.4% Recall through training and evaluation of a custom object detection model.
- Developed Streamlit and Power BI dashboards for real-time monitoring, reporting, and waste analytics.

GitHub Link - Biomedical Waste Segregation Project

RepoMind — AI-powered Code Analysis & Search Platform | *Python, LangGraph, Qdrant Cloud, FastAPI, PostgreSQL, React Flow*

- Developed a semantic search engine (RAG) using LangGraph and Qdrant Cloud for interactive codebase Q&A with source citations.
- Constructed a Graph Database parser to automatically map and visualize complex multi-file repository architecture relationships.
- Built security auditing pipeline scanning repositories with line-by-line recommendation logs.
- Integrated secure GitHub OAuth and background workers for seamless codebase imports.

GitHub Link - RepoMind Project

AI-Based Gas & Smoke Early Warning System | *Python, Random Forest, STM32, UART, Actuators*

- Engineered an Edge ML early warning system utilizing a trained Random Forest classifier.
- Collected and processed a dataset of 28,000+ gas sensor readings for model training.
- Configured bidirectional UART communication between PC inference service and STM32 microcontroller.
- Deployed hardware actuators (exhaust fans, alarms) triggered automatically by UART inference signals.

GitHub Link - Gas & Smoke Early Warning Project

Live Sign Language Transcription System | *Python, TensorFlow, OpenCV, RNN, LSTM*

- Developed a real-time American Sign Language (ASL) transcription system using deep learning techniques.
- Designed an RNN-LSTM architecture for sequential gesture recognition and real-time text generation.
- Implemented OpenCV-based gesture capture and preprocessing pipelines for live inference.

GitHub Link - Sign Language Transcription Project

Technical Skills

Programming Languages: Python, C/C++, SQL
Machine Learning & Deep Learning: Scikit-Learn, TensorFlow, Keras, PyTorch, Deep Learning, Model Evaluation
Computer Vision: OpenCV, YOLOv11, Object Detection, Image Classification, Dataset Annotation, Data Augmentation
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Power BI, Exploratory Data Analysis (EDA)
AI Frameworks & Agentic AI: LangChain, LangGraph, Agentic AI, RAG, Vector Databases
Embedded Systems & IoT: STM32, ESP32, Arduino, Raspberry Pi Pico, UART Communication, MQTT Protocols, Hardware Actuators
Development & Deployment: Streamlit, FastAPI, Git, GitHub, Roboflow, Vercel, Render